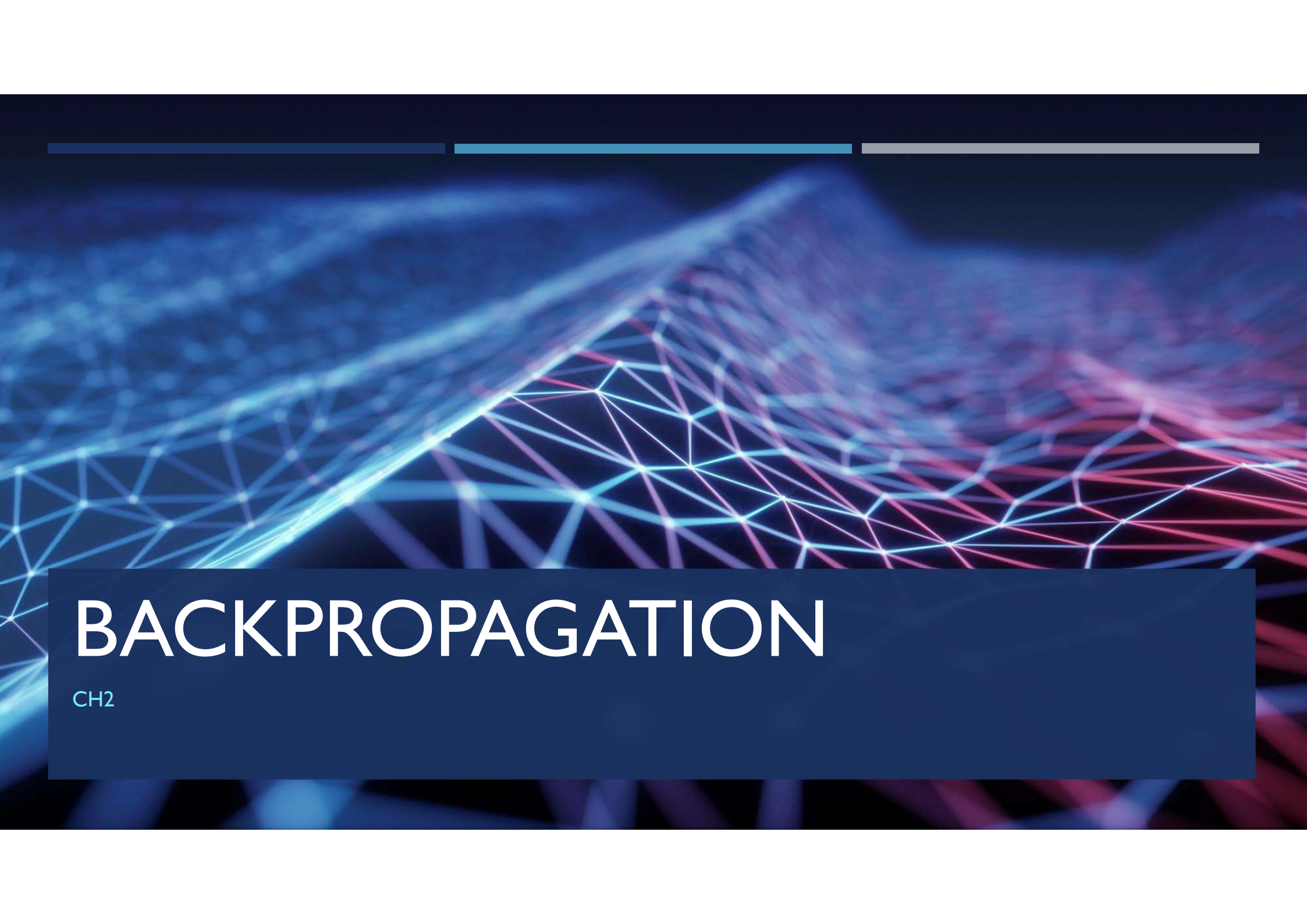
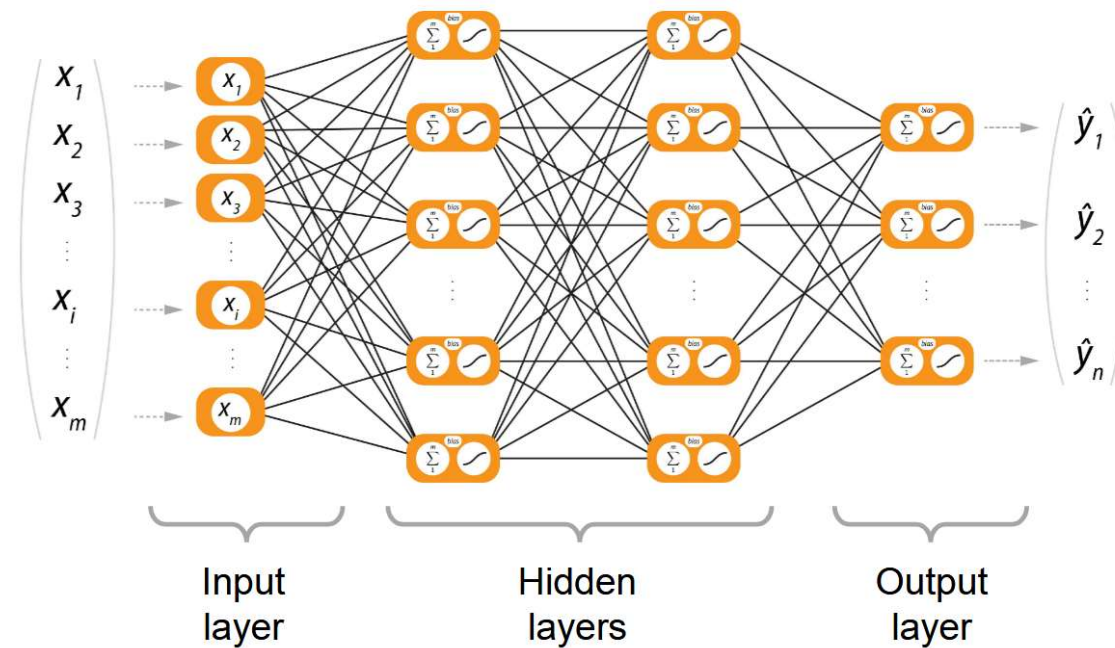




BACKPROPAGATION

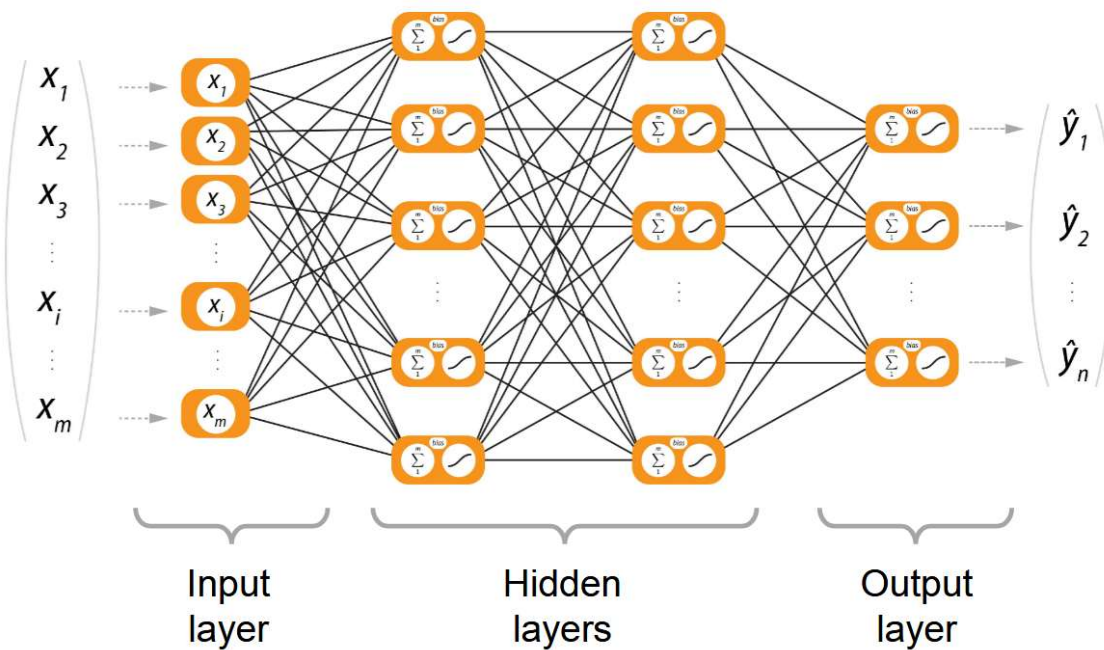
CH2

DEEP MULTI-LAYER PERCEPTRON (MLP) : DEEP NEURAL NETWORKS



LOSS FUNCTION

The loss of our network measure the gap between the predictions and the ground truth



$$\mathcal{L}(\underbrace{f(x^{(i)}; \mathbf{W})}_{\text{Predicted}}, \underbrace{y^{(i)}}_{\text{Actual}})$$

EMPIRICAL LOSS

The empirical loss measures the total loss over our entire dataset.

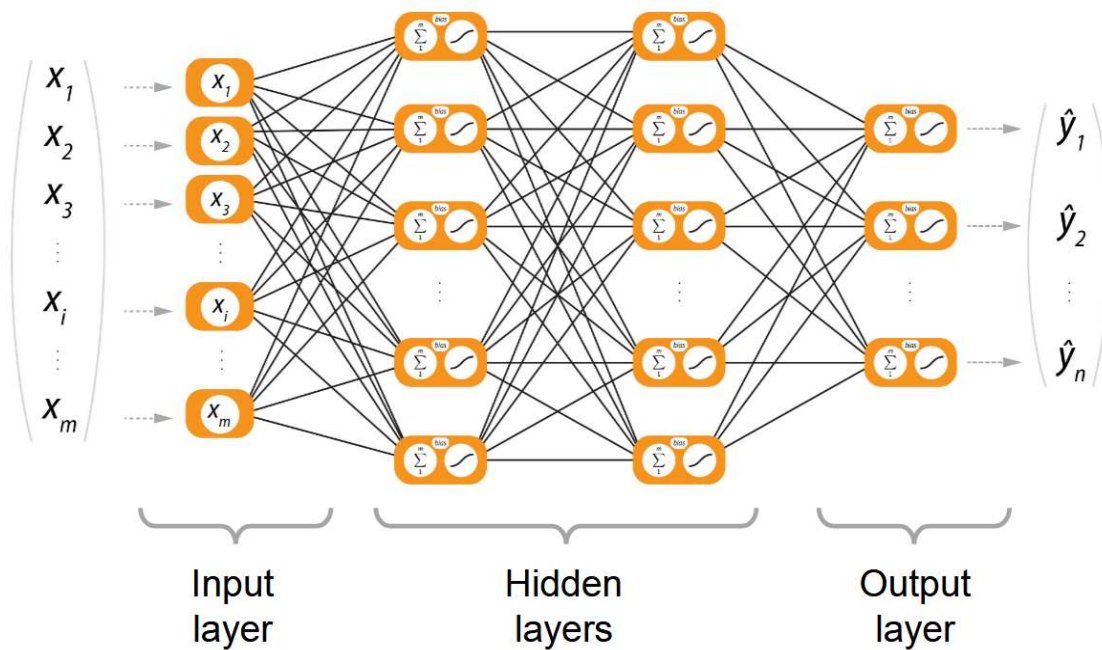
Also known as:

- Objective function
- Cost function
- Empirical Risk


$$J(\mathbf{W}) = \frac{1}{n} \sum_{i=1}^n \mathcal{L}(\underbrace{f(x^{(i)}; \mathbf{W})}_{\text{Predicted}}, \underbrace{y^{(i)}}_{\text{Actual}})$$

LOSS OPTIMIZATION

We want to find the network weights that achieve the lowest loss

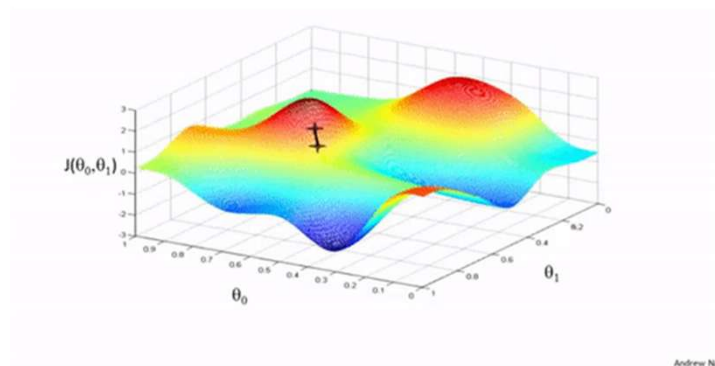
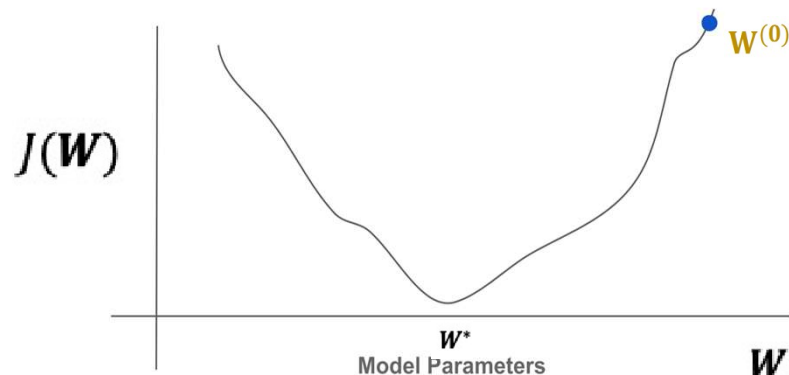


$$\mathbf{W}^* = \operatorname{argmin}_{\mathbf{W}} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f(x^{(i)}; \mathbf{W}), y^{(i)})$$

$$\mathbf{W}^* = \operatorname{argmin}_{\mathbf{W}} J(\mathbf{W})$$

LOSS OPTIMIZATION

We want to move from the initial w_0 to the optimal w^* over the training epochs.



Randomly pick an initial w_0

$$\sim \mathcal{N}(0, \sigma^2)$$

Compute gradient

$$\frac{\partial J(W)}{\partial W}$$

Take small step in opposite direction of gradient : **Gradient-descent step**

$$\text{Update weights, } W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$$

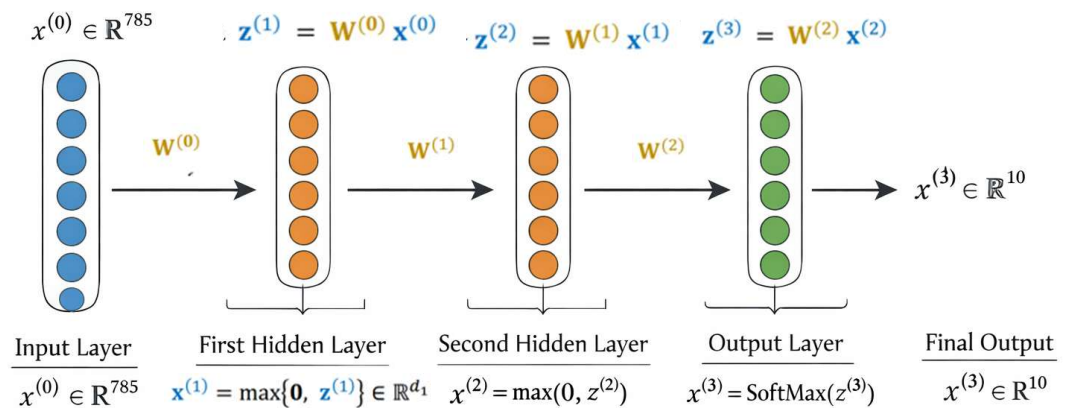
GRADIENT AND BACKPROPAGATION



TRAIN THE NEURAL NETWORK

Define a function $f: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- **Input:** vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
- $\mathbf{x}^{(1)} = \max\{\mathbf{0}, \mathbf{z}^{(1)}\} \in \mathbb{R}^{d_1}$.
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- **Output:** $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.



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- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

Build an optimization model:

$$\underset{\mathbf{W}^{(0)}, \mathbf{W}^{(1)}, \mathbf{W}^{(2)}}{\operatorname{argmin}} \left\{ \frac{1}{n} \sum_{j=1}^n \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j) \right\}$$

E.g., the cross-entropy loss

TRAIN THE NEURAL NETWORK

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- **Output:** $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to solve

$$\underset{\mathbf{W}^{(0)}, \mathbf{W}^{(1)}, \mathbf{W}^{(2)}}{\operatorname{argmin}} \left\{ \frac{1}{n} \sum_{j=1}^n \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j) \right\} ?$$

Stochastic gradient descent (SGD)

TRAIN THE NEURAL NETWORK

Define a function $\mathbf{f}: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
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- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $\mathbf{f}(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to solve

$$\underset{\mathbf{W}^{(0)}, \mathbf{W}^{(1)}, \mathbf{W}^{(2)}}{\operatorname{argmin}} \left\{ \frac{1}{n} \sum_{j=1}^n \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j) \right\} ?$$

Stochastic gradient descent (SGD):

- Randomly pick j from $\{1, 2, \dots, n\}$.
- Compute the stochastic gradient w.r.t. $\mathbf{W}^{(0)}$ at the current iteration $\mathbf{W}_{\text{old}}^{(0)}$:

$$\mathbf{g}_j^{(0)} = \left. \frac{\partial \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(0)}} \right|_{\mathbf{W}^{(0)} = \mathbf{W}_{\text{old}}^{(0)}}$$

- Update $\mathbf{W}^{(0)}$: $\mathbf{W}_{\text{new}}^{(0)} = \mathbf{W}_{\text{old}}^{(0)} - \alpha \mathbf{g}_j^{(0)}$.
- Do the same for $\mathbf{W}^{(1)}$ and $\mathbf{W}^{(2)}$.

BACKPROPAGATION

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- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

BACKPROPAGATION

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- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
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How to compute $\frac{\partial \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)$.

BACKPROPAGATION

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Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

10-dim vector



BACKPROPAGATION

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- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

$\mathbf{z}^{(3)}$ is a function of $\mathbf{z}^{(2)}$ and $\mathbf{W}^{(2)}$.

Apply the chain rule.

BACKPROPAGATION

Define a function $f: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
- $\mathbf{x}^{(1)} = \max\{\mathbf{0}, \mathbf{z}^{(1)}\} \in \mathbb{R}^{d_1}$.
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- $\mathbf{x}^{(2)} = \max\{\mathbf{0}, \mathbf{z}^{(2)}\} \in \mathbb{R}^{d_2}$.
- $\mathbf{z}^{(3)} = \mathbf{W}^{(2)} \mathbf{x}^{(2)} \in \mathbb{R}^{10}$.
- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.

- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

$$\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}} \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$$

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Apply the chain rule.

BACKPROPAGATION

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- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
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- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $\mathbf{f}(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$, $\frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

Then free $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$ from memory.

BACKPROPAGATION

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- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$, $\frac{\partial L}{\partial \mathbf{W}^{(2)}}$ $= \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

Use it to update $\mathbf{W}^{(2)}$ (e.g., by SGD).

BACKPROPAGATION

Define a function $f: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
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How to compute $\frac{\partial \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$, $\boxed{\frac{\partial L}{\partial \mathbf{W}^{(2)}}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

Then free $\frac{\partial L}{\partial \mathbf{W}^{(2)}}$ from memory.

BACKPROPAGATION

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Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.

- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

$$\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}.$$

$$\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{z}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{W}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}.$$

Apply the chain rule again.

BACKPROPAGATION

Define a function $\mathbf{f}: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
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- Output: $\mathbf{f}(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)$.

- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

$$\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}.$$

$$\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{z}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{W}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}.$$

Free $\frac{\partial L}{\partial \mathbf{z}^{(2)}}$ from memory.

BACKPROPAGATION

Define a function $f: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
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- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{z}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}, \quad \boxed{\frac{\partial L}{\partial \mathbf{W}^{(1)}}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{W}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}$.

Use it to update $\mathbf{W}^{(1)}$ (e.g., by SGD).

BACKPROPAGATION

Define a function $f: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

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- $\mathbf{x}^{(2)} = \max\{\mathbf{0}, \mathbf{z}^{(2)}\} \in \mathbb{R}^{d_2}$.
- $\mathbf{z}^{(3)} = \mathbf{W}^{(2)} \mathbf{x}^{(2)} \in \mathbb{R}^{10}$.
- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{z}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}, \quad \boxed{\frac{\partial L}{\partial \mathbf{W}^{(1)}}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{W}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}$.

Free $\frac{\partial L}{\partial \mathbf{W}^{(1)}}$ from memory.

BACKPROPAGATION

Define a function $\mathbf{f}: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
- $\mathbf{x}^{(1)} = \max\{\mathbf{0}, \mathbf{z}^{(1)}\} \in \mathbb{R}^{d_1}$.
- $\mathbf{z}^{(2)} = \mathbf{W}^{(1)} \mathbf{x}^{(1)} \in \mathbb{R}^{d_2}$.
- $\mathbf{x}^{(2)} = \max\{\mathbf{0}, \mathbf{z}^{(2)}\} \in \mathbb{R}^{d_2}$.
- $\mathbf{z}^{(3)} = \mathbf{W}^{(2)} \mathbf{x}^{(2)} \in \mathbb{R}^{10}$.
- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $\mathbf{f}(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)$.

- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.

$$\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}.$$

$$\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{z}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{W}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}.$$

$$\frac{\partial L}{\partial \mathbf{W}^{(0)}} = \frac{\partial \mathbf{z}^{(1)}}{\partial \mathbf{W}^{(0)}} \frac{\partial L}{\partial \mathbf{z}^{(1)}}.$$

Apply the chain rule again.

BACKPROPAGATION

Define a function $f: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
- $\mathbf{x}^{(1)} = \max\{\mathbf{0}, \mathbf{z}^{(1)}\} \in \mathbb{R}^{d_1}$.
- $\mathbf{z}^{(2)} = \mathbf{W}^{(1)} \mathbf{x}^{(1)} \in \mathbb{R}^{d_2}$.
- $\mathbf{x}^{(2)} = \max\{\mathbf{0}, \mathbf{z}^{(2)}\} \in \mathbb{R}^{d_2}$.
- $\mathbf{z}^{(3)} = \mathbf{W}^{(2)} \mathbf{x}^{(2)} \in \mathbb{R}^{10}$.
- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(f(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}.$
- $\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{z}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}},$
- $\frac{\partial L}{\partial \mathbf{W}^{(0)}} = \frac{\partial \mathbf{z}^{(1)}}{\partial \mathbf{W}^{(0)}} \frac{\partial L}{\partial \mathbf{z}^{(1)}}.$

Free $\frac{\partial L}{\partial \mathbf{z}^{(1)}}$ from memory.

BACKPROPAGATION

Define a function $f: \mathbb{R}^{785} \mapsto \mathbb{R}^{10}$:

- Input: vector $\mathbf{x}^{(0)} \in \mathbb{R}^{785}$.
- $\mathbf{z}^{(1)} = \mathbf{W}^{(0)} \mathbf{x}^{(0)} \in \mathbb{R}^{d_1}$.
- $\mathbf{x}^{(1)} = \max\{\mathbf{0}, \mathbf{z}^{(1)}\} \in \mathbb{R}^{d_1}$.
- $\mathbf{z}^{(2)} = \mathbf{W}^{(1)} \mathbf{x}^{(1)} \in \mathbb{R}^{d_2}$.
- $\mathbf{x}^{(2)} = \max\{\mathbf{0}, \mathbf{z}^{(2)}\} \in \mathbb{R}^{d_2}$.
- $\mathbf{z}^{(3)} = \mathbf{W}^{(2)} \mathbf{x}^{(2)} \in \mathbb{R}^{10}$.
- $\mathbf{x}^{(3)} = \text{SoftMax}(\mathbf{z}^{(3)}) \in \mathbb{R}^{10}$.
- Output: $f(\mathbf{x}^{(0)}) = \mathbf{x}^{(3)}$.

How to compute $\frac{\partial \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)}{\partial \mathbf{W}^{(k)}}$?

Backpropagation:

- Denote $L = \text{Loss}(\mathbf{f}(\mathbf{x}_j), \mathbf{y}_j)$.
- Compute $\frac{\partial L}{\partial \mathbf{z}^{(3)}}$.
- $\frac{\partial L}{\partial \mathbf{z}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{z}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(2)}} = \frac{\partial \mathbf{z}^{(3)}}{\partial \mathbf{W}^{(2)}} \frac{\partial L}{\partial \mathbf{z}^{(3)}}.$
- $\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{z}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}, \quad \frac{\partial L}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathbf{z}^{(2)}}{\partial \mathbf{W}^{(1)}} \frac{\partial L}{\partial \mathbf{z}^{(2)}}.$
- $\frac{\partial L}{\partial \mathbf{W}^{(0)}} = \frac{\partial \mathbf{z}^{(1)}}{\partial \mathbf{W}^{(0)}} \frac{\partial L}{\partial \mathbf{z}^{(1)}}.$ Use it to update $\mathbf{W}^{(0)}$.

BACKPROPAGATION: EXAMPLE



ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- **Input:** scalar $x^{(0)}$.
- $z^{(1)} = w^{(0)} x^{(0)}$.
- $x^{(1)} = \max\{0, z^{(1)}\}$.
- $z^{(2)} = w^{(1)} x^{(1)}$.
- $x^{(2)} = \max\{0, z^{(2)}\}$.
- $z^{(3)} = w^{(2)} x^{(2)}$.
- **Output:** $f(x^{(0)}) = z^{(3)}$.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- **Input:** scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - **Output:** $f(x^{(0)}) = z^{(3)}$.

Random sampling:

- Randomly sample j from $\{1, 2, \dots, n\}$.

Forward pass (input $\rightarrow$ output):

- Take x_j as input ($x^{(0)} = x_j$).
- Compute each layer $z^{(1)}, x^{(1)}, z^{(2)}, x^{(2)}, z^{(3)}$.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- **Input:** scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - **Output:** $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$



- **Input:** scalar $x^{(0)}$.
- $z^{(1)} = w^{(0)} x^{(0)}$.
- $x^{(1)} = \max\{0, z^{(1)}\}$.
- $z^{(2)} = w^{(1)} x^{(1)}$.
- $x^{(2)} = \max\{0, z^{(2)}\}$.
- $z^{(3)} = w^{(2)} x^{(2)}$.
- **Output:** $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2 = \frac{1}{2} (z^{(3)} - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- **Input:** scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - **Output:** $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = \boxed{z^{(3)}} - y_j$.

The value of $z^{(3)}$ is known
(after the forward pass).

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- **Input:** scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - **Output:** $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.

The value of $z^{(3)}$ is known
(after the forward pass).

Thus the value of $\frac{\partial L}{\partial z^{(3)}}$ is known.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \boxed{\frac{\partial z^{(3)}}{\partial z^{(2)}}} \frac{\partial L}{\partial z^{(3)}}$

$$\frac{\partial z^{(3)}}{\partial x^{(2)}} = w^{(2)}, \quad \frac{\partial x^{(2)}}{\partial z^{(2)}} = \begin{cases} 1, & \text{if } z^{(2)} > 0; \\ 0, & \text{else.} \end{cases}$$

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$

$$\frac{\partial z^{(3)}}{\partial x^{(2)}} = w^{(2)}, \quad \frac{\partial x^{(2)}}{\partial z^{(2)}} = \begin{cases} 1, & \text{if } z^{(2)} > 0; \\ 0, & \text{else.} \end{cases}$$


$$\frac{\partial z^{(3)}}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial x^{(2)}} \frac{\partial x^{(2)}}{\partial z^{(2)}} = \begin{cases} w^{(2)}, & \text{if } z^{(2)} > 0; \\ 0, & \text{else.} \end{cases}$$

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.

$$\frac{\partial z^{(3)}}{\partial w^{(2)}} = x^{(2)}.$$

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.

Free $\frac{\partial L}{\partial z^{(3)}}$ from memory.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\boxed{\frac{\partial L}{\partial w^{(2)}}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.

Update $w^{(2)}$: $w^{(2)} \leftarrow w^{(2)} - \alpha \frac{\partial L}{\partial w^{(2)}}$.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.

Free $\frac{\partial z^{(3)}}{\partial w^{(2)}}$ from memory.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$



- Input: scalar $x^{(0)}$.
- $z^{(1)} = w^{(0)} x^{(0)}$.
- $x^{(1)} = \max\{0, z^{(1)}\}$.
- $z^{(2)} = w^{(1)} x^{(1)}$.
- $x^{(2)} = \max\{0, z^{(2)}\}$.
- $z^{(3)} = w^{(2)} x^{(2)}$.
- Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)}}$

$$\frac{\partial z^{(2)}}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial x^{(1)}} \frac{\partial x^{(1)}}{\partial z^{(1)}} = \begin{cases} w^{(1)}, & \text{if } z^{(1)} > 0; \\ 0, & \text{else.} \end{cases}$$

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2}(f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)}}$, $\frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)}}$.

$$\frac{\partial z^{(2)}}{\partial w^{(1)}} = x^{(1)}.$$

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)'}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)'}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)'}}$, $\frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)'}}$.

Free $\frac{\partial L}{\partial z^{(2)}}$ from memory.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)}}$, $\frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)}}$.

Update $w^{(1)}$: $w^{(1)} \leftarrow w^{(1)} - \alpha \frac{\partial L}{\partial w^{(1)}}$.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$



- Input: scalar $x^{(0)}$.
- $z^{(1)} = w^{(0)} x^{(0)}$.
- $x^{(1)} = \max\{0, z^{(1)}\}$.
- $z^{(2)} = w^{(1)} x^{(1)}$.
- $x^{(2)} = \max\{0, z^{(2)}\}$.
- $z^{(3)} = w^{(2)} x^{(2)}$.
- Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)}}$, $\frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)}}$.

Free $\frac{\partial L}{\partial w^{(1)}}$ from memory.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$



- Input: scalar $x^{(0)}$.
- $z^{(1)} = w^{(0)} x^{(0)}$.
- $x^{(1)} = \max\{0, z^{(1)}\}$.
- $z^{(2)} = w^{(1)} x^{(1)}$.
- $x^{(2)} = \max\{0, z^{(2)}\}$.
- $z^{(3)} = w^{(2)} x^{(2)}$.
- Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)}}$, $\frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)}}$.
- $\frac{\partial L}{\partial w^{(0)}} = \frac{\partial z^{(1)}}{\partial w^{(0)}} \frac{\partial L}{\partial z^{(1)}}$.

$$\frac{\partial z^{(1)}}{\partial w^{(0)}} = x^{(0)}.$$

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)}}$ $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)}}$ $\frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)}}$.
- $\frac{\partial L}{\partial w^{(0)}} = \frac{\partial z^{(1)}}{\partial w^{(0)}} \frac{\partial L}{\partial z^{(1)}}$.

Free $\frac{\partial L}{\partial z^{(1)}}$ from memory.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

- 
- Input: scalar $x^{(0)}$.
 - $z^{(1)} = w^{(0)} x^{(0)}$.
 - $x^{(1)} = \max\{0, z^{(1)}\}$.
 - $z^{(2)} = w^{(1)} x^{(1)}$.
 - $x^{(2)} = \max\{0, z^{(2)}\}$.
 - $z^{(3)} = w^{(2)} x^{(2)}$.
 - Output: $f(x^{(0)}) = z^{(3)}$.

Backpropagation:

- Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.
- $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.
- $\frac{\partial L}{\partial z^{(2)}} = \frac{\partial z^{(3)}}{\partial z^{(2)}} \frac{\partial L}{\partial z^{(3)'}}$, $\frac{\partial L}{\partial w^{(2)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)'}}$.
- $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)'}}$, $\frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)'}}$.
- $\frac{\partial L}{\partial w^{(0)}} = \frac{\partial z^{(1)}}{\partial w^{(0)}} \frac{\partial L}{\partial z^{(1)'}}$.

Update $w^{(0)}$: $w^{(0)} \leftarrow w^{(0)} - \alpha \frac{\partial L}{\partial w^{(0)'}}$.

ID EXAMPLE

Define a function $f: \mathbb{R} \mapsto \mathbb{R}$

Backpropagation:

One iteration:

1. Randomly sample j from $\{1, 2, \dots, n\}$.

2. **Forward pass:** take x_j as input ($x^{(0)} = x_j$), compute each layer $z^{(1)}, x^{(1)}, z^{(2)}, x^{(2)}, z^{(3)}$.

3. **Backward pass:**

- Compute the derivatives $\frac{\partial L}{\partial z^{(3)}}, \frac{\partial L}{\partial w^{(2)}}, \frac{\partial L}{\partial z^{(2)}}, \frac{\partial L}{\partial w^{(1)}}, \frac{\partial L}{\partial z^{(1)}}, \frac{\partial L}{\partial w^{(0)}}$.
- Update $w^{(k)}$ using $\frac{\partial L}{\partial w^{(k)}}$.

• Loss: $L = \frac{1}{2} (f(x_j) - y_j)^2$.

• $\frac{\partial L}{\partial z^{(3)}} = z^{(3)} - y_j$.

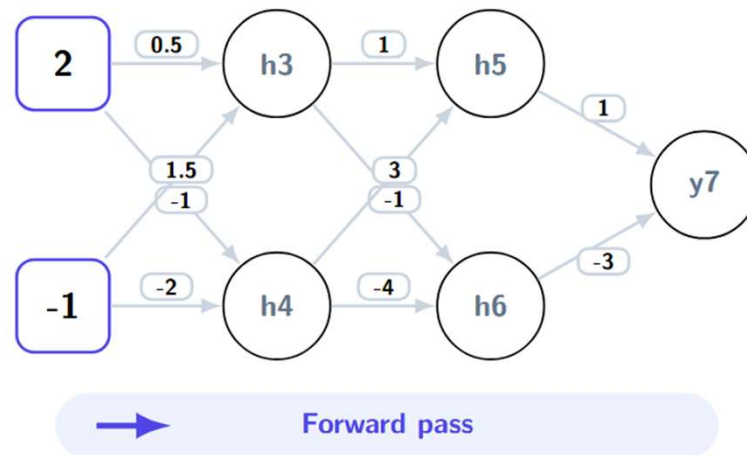
• $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(3)}}{\partial w^{(2)}} \frac{\partial L}{\partial z^{(3)}}$.

• $\frac{\partial L}{\partial z^{(1)}} = \frac{\partial z^{(2)}}{\partial z^{(1)}} \frac{\partial L}{\partial z^{(2)}} \frac{\partial L}{\partial w^{(1)}} = \frac{\partial z^{(2)}}{\partial w^{(1)}} \frac{\partial L}{\partial z^{(2)}}$.

BACKPROPAGATION: EXAMPLE 2



EXAMPLE – NETWORK AND WEIGHTS



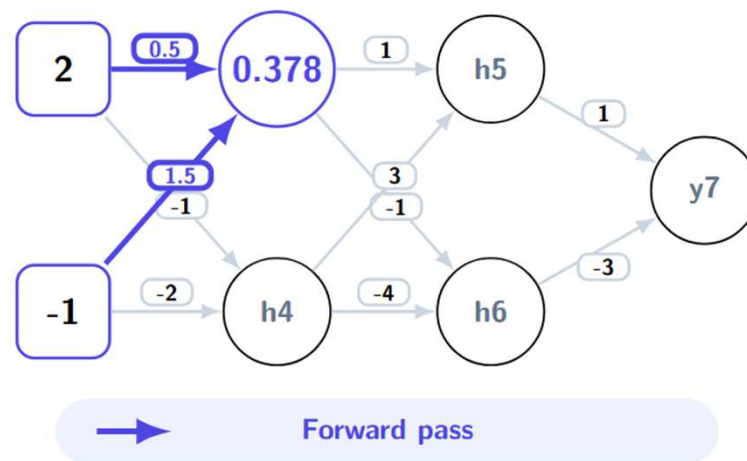
Forward pass

$$a_k = g\left(\sum_j w_{j,k} \cdot a_j\right)$$

Sigmoid

$$g(x) = \frac{1}{1 + e^{-x}}$$

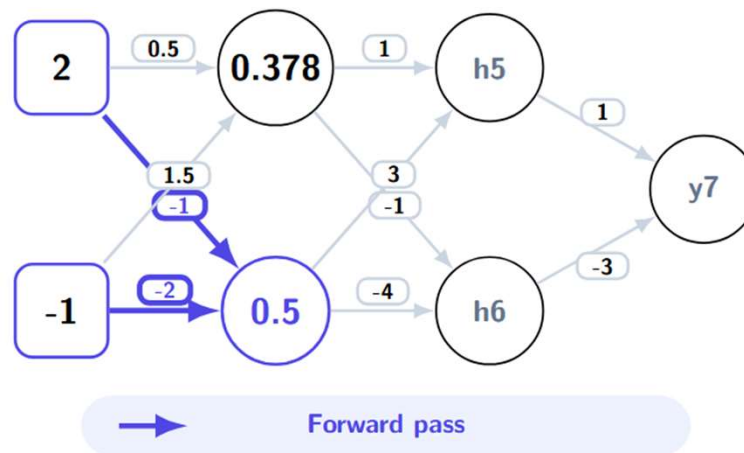
FORWARD PASS – NEURON 3



Computation

$$\begin{aligned} a_3 &= g(0.5 \cdot 2 + 1.5 \cdot (-1)) \\ &= g(-0.5) \\ &= 0.378 \end{aligned}$$

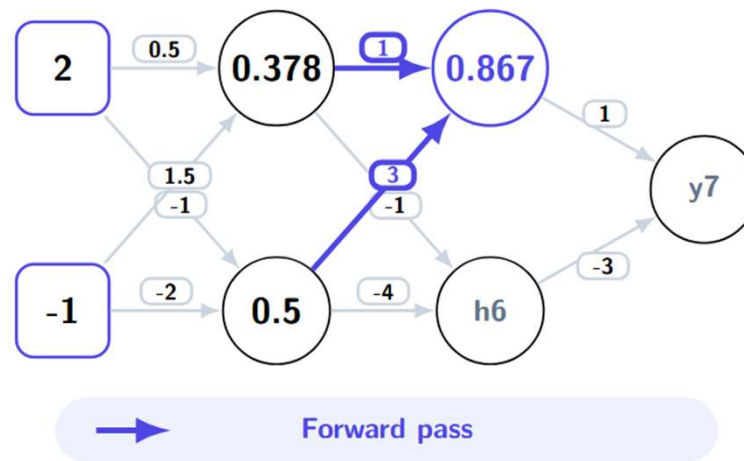
FORWARD PASS – NEURON 4



Computation

$$\begin{aligned} a_4 &= g((-1) \cdot 2 + (-2) \cdot (-1)) \\ &= g(0) \\ &= 0.5 \end{aligned}$$

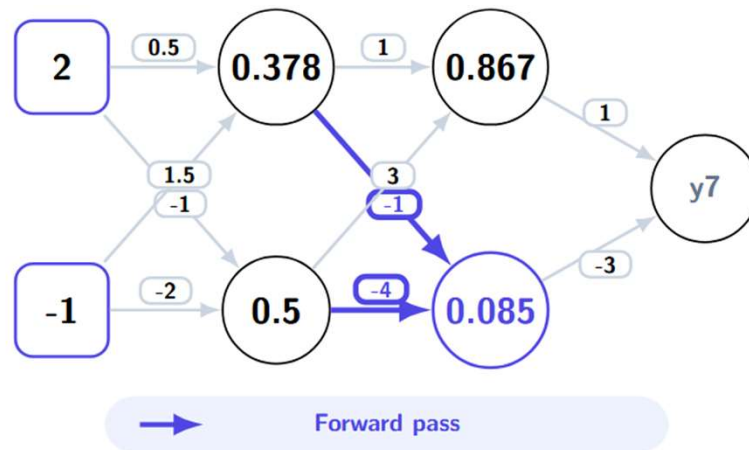
FORWARD PASS – NEURON 5



Computation

$$\begin{aligned} a_5 &= g(1 \cdot a_3 + 3 \cdot a_4) \\ &= g(1 \cdot 0.378 + 3 \cdot 0.5) \\ &= g(1.878) \\ &= 0.867 \end{aligned}$$

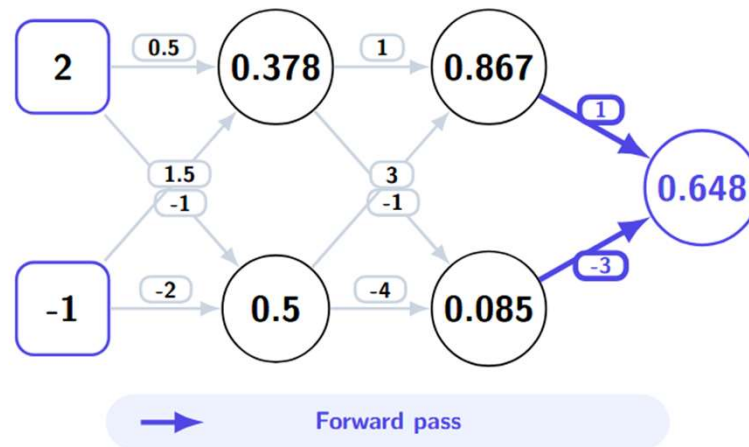
FORWARD PASS – NEURON 6



Computation

$$\begin{aligned} a_6 &= g((-1) \cdot a_3 + (-4) \cdot a_4) \\ &= g((-1) \cdot 0.378 + (-4) \cdot 0.5) \\ &= g(-2.378) \\ &= 0.085 \end{aligned}$$

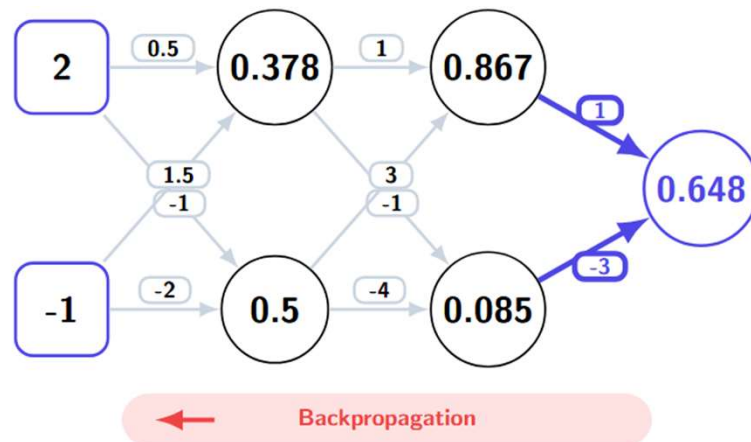
FORWARD PASS – OUTPUT



Computation

$$\begin{aligned}a_7 &= g(1 \cdot a_5 + (-3) \cdot a_6) \\&= g(1 \cdot 0.867 + (-3) \cdot 0.085) \\&= g(0.612) \\&= 0.648\end{aligned}$$

BACKPROPAGATION – PRINCIPLE



Rules

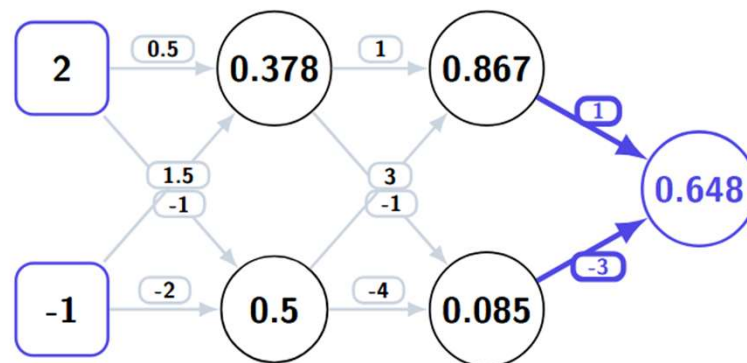
$$\mathcal{L}(y, a) = -(y \log(a) + (1-y) \log(1-a))$$

$$\delta_7 = y - a_7$$

$$\delta_j = a_j(1 - a_j) \cdot \sum_k w_{j,k} \delta_k$$

(since $g'(a) = a(1 - a)$)

OUTPUT DELTA



Computation

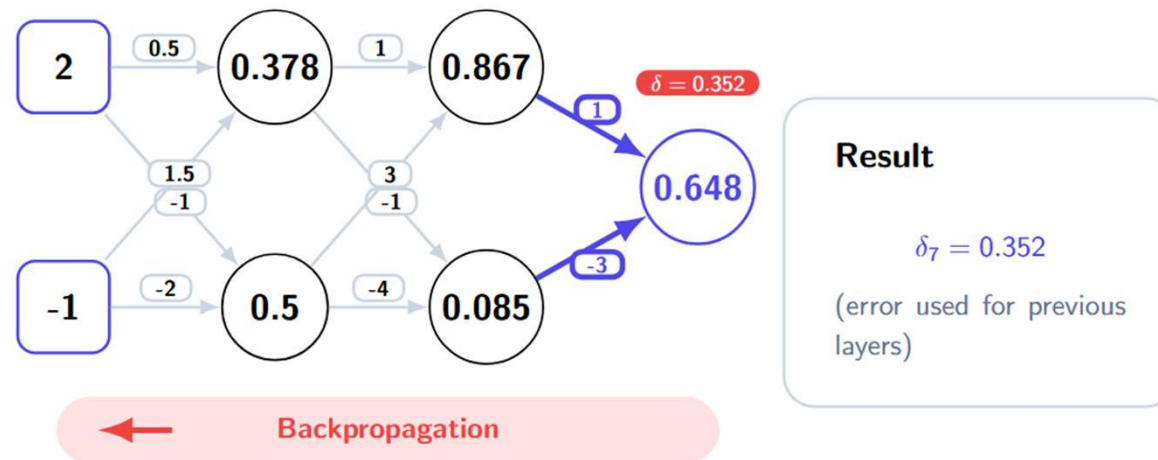
$$\delta_7 = y - a_7$$

$$\delta_7 = 1 - 0.648$$

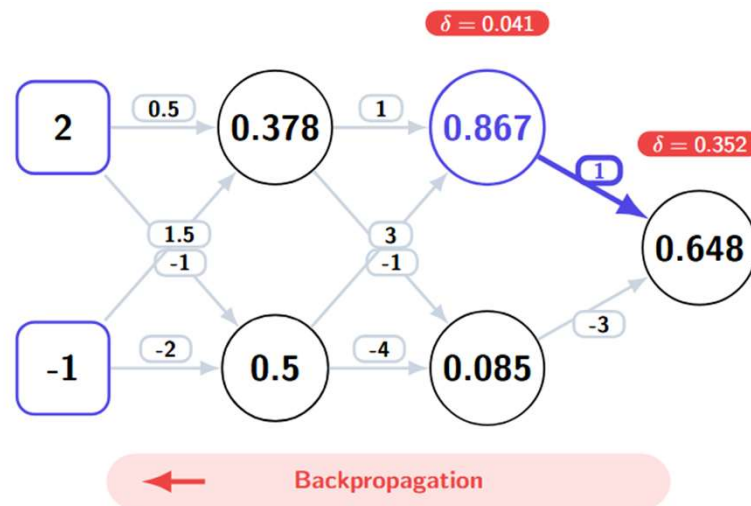


Backpropagation

OUTPUT DELTA



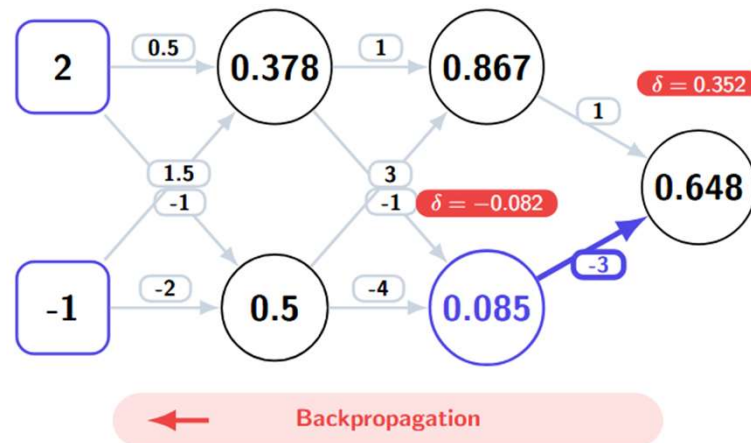
DELTA OF NEURON 5



Computation

$$\begin{aligned}\delta_5 &= a_5(1 - a_5) \cdot w_{5,7} \cdot \delta_7 \\ &= 0.867(1 - 0.867) \cdot 1 \cdot 0.352 \\ &= 0.041\end{aligned}$$

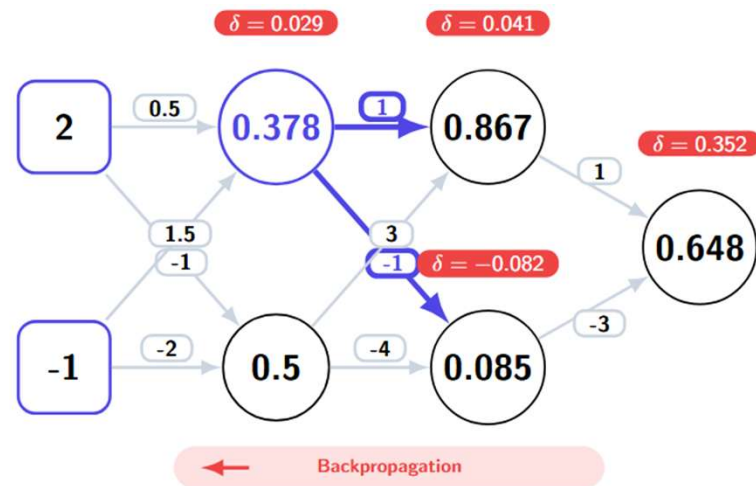
DELTA OF NEURON 6



Computation

$$\begin{aligned}\delta_6 &= a_6(1 - a_6) \cdot w_{6,7} \cdot \delta_7 \\ &= 0.085(1 - 0.085) \cdot (-3) \cdot 0.352 \\ &= -0.082\end{aligned}$$

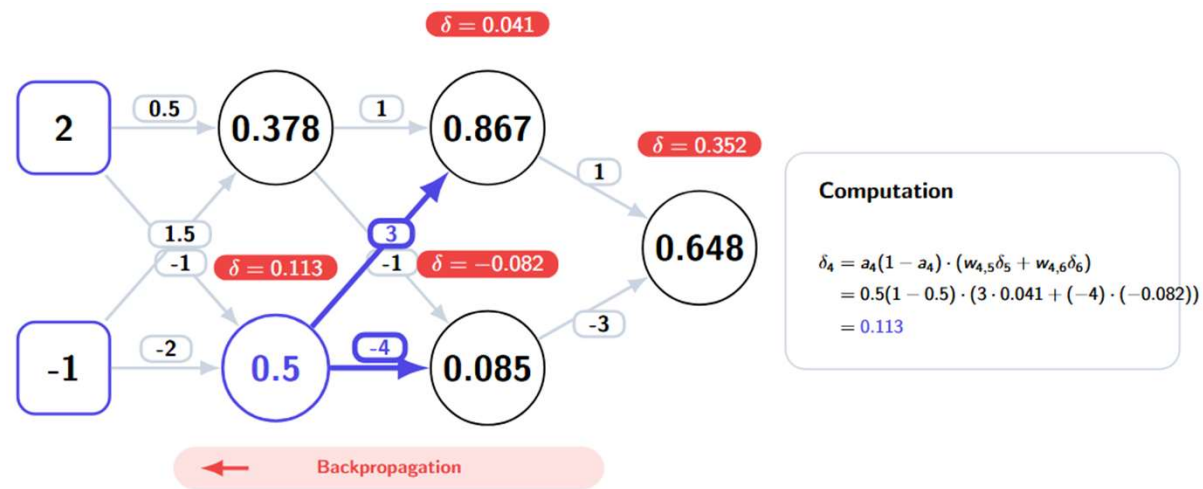
DELTA OF NEURON 3



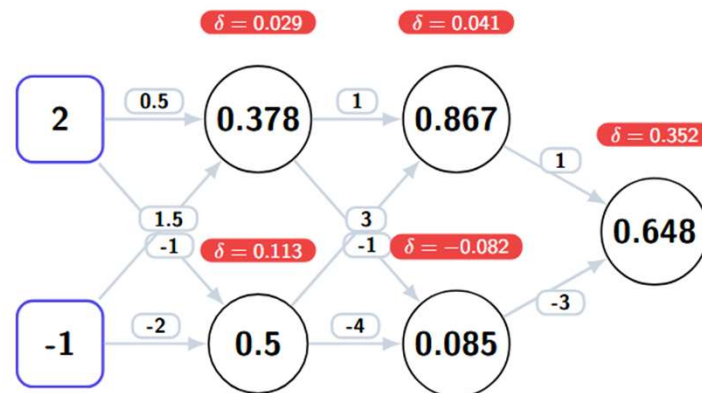
Computation

$$\begin{aligned}\delta_3 &= a_3(1 - a_3) \cdot (w_{3,5}\delta_5 + w_{3,6}\delta_6) \\ &= 0.378(1 - 0.378) \cdot (1 \cdot 0.041 + (-1) \cdot (-0.082)) \\ &= 0.029\end{aligned}$$

DELTA OF NEURON 4



WEIGHT UPDATE



New weights

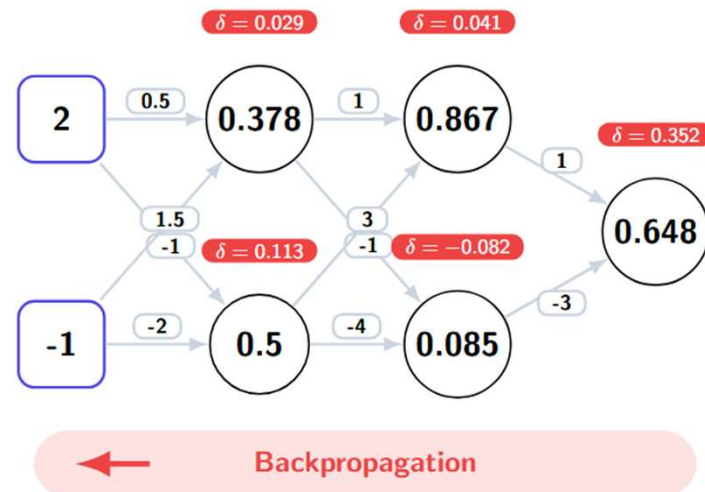
General rule:

$$w_{i,j}^{new} = w_{i,j}^{old} + \alpha a_i \delta_j$$



Backpropagation

WEIGHT UPDATE



New weights

After one iteration with $\alpha = 0.1$

$w_{1,3}$	0.5	$\rightarrow$	0.506
$w_{1,4}$	-1	$\rightarrow$	-0.977
$w_{2,3}$	1.5	$\rightarrow$	1.497
$w_{2,4}$	-2	$\rightarrow$	-2.011
$w_{3,5}$	1	$\rightarrow$	1.002
$w_{3,6}$	-1	$\rightarrow$	-1.003
$w_{4,5}$	3	$\rightarrow$	3.002
$w_{4,6}$	-4	$\rightarrow$	-4.004
$w_{5,7}$	1	$\rightarrow$	1.031
$w_{6,7}$	-3	$\rightarrow$	-2.997



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